

Recap

ERM

$$\hat{h}_S \in \operatorname{argmin}_{h \in H} \underbrace{L_S(h)}_{:= \frac{1}{m} \sum_{i=1}^m \ell(h, z_i)} \\ S = \{z_1, \dots, z_m\}$$

Question: would like that $L_D(\hat{h}_S)$ is small with HIGH PROBAB.? (with respect to the choice of S)

Warning: we have seen a simple example where

$$L_S(\hat{h}_S) = 0 \quad \text{but} \quad L_D(\hat{h}_S) = 1/2$$

Goal of this lecture: ...

We want to guarantee that $L_D(\hat{h}_S) - \underbrace{L_S(\hat{h}_S)}_{=0}$ is small
generalization gap $\rightarrow$ simplifying assumption

(1) Want $L_D(\hat{h}_S) \leq \epsilon$

(2) $\mathbb{P}_{S \sim D^m} [L_D(\hat{h}_S) \leq \epsilon] \geq 1 - \delta$

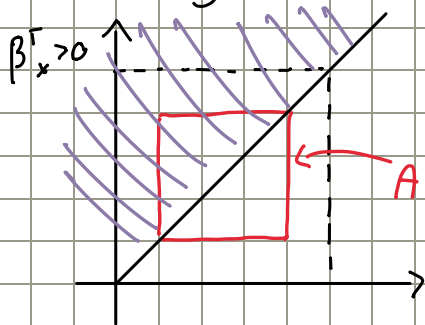
(segno: quello che ho appena detto, minuto 19:30)

Can I always choose ϵ arbitrarily small?

Assumption [Realizability]: indeed unrealistic

$$\exists h \in H \text{ s.t. } L_D(h) = 0$$

Reminding lesson 2 example



$$D \sim \begin{cases} x_1 \sim U([0,1] \times [0,1]) \\ y = \begin{cases} 1 & x_i \in A \\ 0 & x_i \notin A \end{cases} \end{cases}$$

$$h_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

(1) All possible functions from $\mathbb{R}^2 \rightarrow \{0,1\}$

$$\mathbb{E}_D[\ell(h_A, z)] = L_D(h_A) = \mathbb{P}_D[\ell(h_A, z) = 1] = \mathbb{P}_D[h_A(x) \neq y] = 0$$

$$(2) H = \left\{ h_\beta(x) = \begin{cases} 1 & \beta_{Tx} > 0 \\ 0 & \beta_{Tx} \leq 0 \end{cases}, \beta \in \mathbb{R}^2 \right\}$$

PAC Learning (Probably Approximate Correct) (Simple)

$$(1) \text{ Want } L_D(\hat{h}_S) \leq \epsilon$$

$$(2) \mathbb{P}_{D \sim D} [L_D(\hat{h}_S) \leq \epsilon] \geq 1 - \delta$$

Definition [PAC LEARNABILITY] (SIMPLE)

$$\text{ASS: } \exists f(\cdot) \text{ s.t. } y = f(x)$$

$$\exists h \in H \text{ s.t. } L_D(h) = 0 \text{ [realizability]}$$

The class H is PAC LEARNABLE if $\exists m_H(\epsilon, \delta) \in \mathbb{N}$

$\exists$ a learning algorithm $A(S) \quad S \rightarrow h_S \in H$ s.t. $\forall \epsilon, \delta \rightarrow h_S \in H$
s.t. $\forall \epsilon, \delta \in (0, 1)^2, \forall D \forall f$ then $\forall m \geq m_H(\epsilon, \delta)$
 $IP[L_D(A(S)) \leq \epsilon] \geq 1 - \delta$

Definition: $m_H(\epsilon, \delta)$

$m_H(\epsilon, \delta)$ is called the sample complexity of H

$$IP[L_D(A(S)) > \epsilon] < \delta$$

THEOREM

Assume $H = \{h_1, \dots, h_{|H|}\}$
└ cardinality of the class

Then if $m \geq \frac{1}{\epsilon} \log\left(\frac{|H|}{\delta}\right)$

$\exists A(S)$ s.t. $P[L_D(A(S)) > \epsilon] < \delta$

There are many unrealistic assumptions

PROOF:

(1) set of bad hypothesis $H_B \subset H$

$$H_B = \{h \in H \text{ s.t. } L_D(h) > \epsilon\}$$

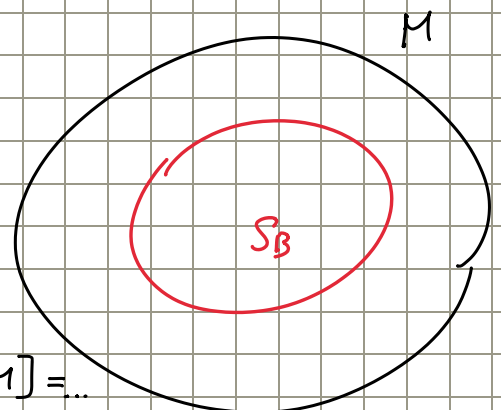
(2) Misleading examples M

$$M = \{S : \exists h \in H_B \quad L_S(h) = 0\}$$

Now, take $\hat{h}_S = \arg\min_{h \in H} L_S(h)$

$$IP_{S \sim D}[L_D(\hat{h}_S) > \epsilon]$$

$$IP[S \text{ s.t. } L_D(\hat{h}_S) > \epsilon] \leq P[M] = \dots$$



S_{Bad}

We want to convince us that $S_0 \subseteq M$

Remark: under realizability assumption, $(\exists h \ L_0(h) = 0)$
 $\Rightarrow L_S(h_S) = 0$ w.p. 1.

$$\dots = P \left[\underbrace{S \ \exists h \in H_0 \text{ s.t. } L_S(h) = 0}_{\bigcup_{h \in H_0} \{S \text{ s.t. } L_S(h) = 0\}} \right] \leq \sum_{h \in H_0} P[S \text{ s.t. } L_S(h) = 0]$$

$$P[A \cup B] \leq P[A] + P[B]$$

Missing here, choose a Perloff